1. Objectives:

1. To understand the basic structure and syntax of a Java program.
2. To implement a simple Java program using package, class, and main method.

2. Introduction:

Java is a high-level, object-oriented programming language developed by Sun Microsystems in 1995 and later acquired by Oracle Corporation. Java supports OOP concepts to create modular and reusable code. It is platform-independent, meaning programs written in Java can run on any system that has a Java Virtual Machine (JVM). Java ensures reliability and security through strong memory management and exception handling.

A Java program is organized into classes. Every application must contain at least one class, and the execution of the program starts from the main() method. The main() method is defined as " public static void main(String[] args) ". In here:

- **public** allows access from anywhere.
- **static** means the method belongs to the class.
- **void** indicates that the method does not return any value.
- **String[] args** is used to pass command-line arguments.

// A Java program to print Hello World!	This is a single-line comment
public class HelloWorld	Defines a class named HelloWorld.
{	There is an Opening Bracket for Class
public static void main(String[] args)	This is the main method, the entry point of any Java program.
{	There is an Opening Bracket for Method
System.out.println("Hello World!")	Output Statement prints "Hello World!" to the console.
}	There is a Closing Bracket for Method
}	There is a Closing Bracket for Class

Fig:1.1: Java Syntax

The package keyword is used to group related classes together. It helps in organizing large projects. A project in Java contains source files, libraries, and other necessary resources. A class is a blueprint for creating objects and defines the structure and behavior of the program. The statement System.out.println() is used to print output on the console and automatically moves the cursor to a new line after printing.

3. Procedure

Experiment 1: Introduction to Java

Algorithm:

1. Start the program.
2. Declare the package name
3. Define a public class name
4. Declare the main() method as the entry point of the program.
5. Use System.out.println() to print the statements
6. End the program.

4. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Visual Studio Code 1.109

5. Experiment Implementation:

Experiment 1: Introduction to Java

```
package MyLab;

public class MyClass {
    public static void main(String[] args) {
        System.out.println("I'm Rezwan");
        System.out.println("ID: 0802510205101024");
        System.err.println("Department: CSE");
        System.out.println("Level 2 Term 1");
        System.out.println("Corse Code: 2108");
    }

}
```

Table 3.1: Experiment 1 output

```
PS D:\BAUST CSE\Java Programing\1st_Lab>  & 'C:\Program Files\Java\jdk-25
.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMes
sages' '-cp' 'D:\BAUST CSE\Java Programing\1st_Lab\bin' 'MyLab.MyClass'
I'm Rezwan
ID: 0802510205101024
Department: CSE
Level 2 Term 1
Corse Code: 2108
PS D:\BAUST CSE\Java Programing\1st_Lab>
```

6. Discussion & Conclusion:

In this lab we have learned about Java, a basic Java program was implemented to understand the structure of a Java application. We have learned the concepts of package declaration, class definition, and the main method. The use of String[] args and System.out.println() shows us how output is displayed in a new line in Java. This lab helped us to learn the foundational knowledge necessary for developing Java programs in the future.

7. Reference:

1. https://www.geeksforgeeks.org/java/java/ accessed at 9.25 PM on 20th February 2026.
2. https://www.geeksforgeeks.org/java/introduction-to-java/ accessed at 4.05 PM on 21st February 2026.